Power Series and Interval of Convergence

ANSWER KEY

Finding the radius of convergence

1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.
$\lim_{n\to\infty}\left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| = \lim_{n\to\infty}\frac{n|x|}{n+1} = |x|$. Converges when $|x| < 1$: radius of convergence $R = 1$.

2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
$\lim_{n\to\infty}\left|\frac{x^{n+1}/(n+1)!}{x^n/n!}\right| = \lim_{n\to\infty}\frac{|x|}{n+1} = 0 < 1$ for all x. Radius of convergence $R = \infty$ (converges everywhere).

Finding the interval of convergence (testing endpoints)

3 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.
At $x = 1$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges (alternating harmonic, by AST). Interval of convergence: $[-1, 1)$.

4 Find the interval of convergence of $\sum \frac{(x-2)^n}{n^2}$.
Ratio Test: $\lim\left|\frac{(x-2)^{n+1}/(n+1)^2}{(x-2)^n/n^2}\right| = |x-2| < 1 \Rightarrow 1 < x < 3$. At $x = 1$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely. At $x = 3$: $\sum \frac{1}{n^2}$ converges. Interval: $[1, 3]$.

Representing functions as power series

5 Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$, valid for $|x| < 1$, i.e. interval of convergence $(-1, 1)$.

6 Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.
Substitute $x \to -x^2$: $\frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

Differentiating and integrating power series

7 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.
$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} nx^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} = 1 + 2x + 3x^2 + \cdots$.

8 Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.
$\tan^{-1} x = \int \frac{1}{1+x^2}\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$. Since $\tan^{-1} 0 = 0$, $C = 0$: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$.

More radius/interval of convergence practice

9 Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n!\, x^n$.
$\lim_{n\to\infty}\left|\frac{(n+1)!\, x^{n+1}}{n!\, x^n}\right| = \lim_{n\to\infty}(n+1)|x| = \infty$ for any $x \neq 0$. Converges only at $x = 0$: $R = 0$, interval $\{0\}$.

10 Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.
$\lim_{n\to\infty}\left|\frac{(x+1)^{n+1}/[(n+1)3^{n+1}]}{(x+1)^n/(n3^n)}\right| = \frac{|x+1|}{3} < 1 \Rightarrow |x+1| < 3 \Rightarrow -4 < x < 2$; $R = 3$. At $x = 2$: $\sum \frac{(-1)^n}{n}$ converges. At $x = -4$: $\sum \frac{1}{n}$ diverges. Interval: $(-4, 2]$.

Visualizing a power series approximation

11 The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.

The two curves nearly overlap near $x = 0$ and diverge visibly as $x \to \pm 0.8$ (closer to the boundary of the interval of convergence), since the partial sum's approximation quality worsens near the edges of convergence.

Term-by-term operations preserve the radius of convergence

12 State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

Differentiating or integrating a power series term by term does not change its radius of convergence R (though convergence AT the endpoints $x = \pm R$ may change).

13 Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

By the term-by-term differentiation theorem, the radius of convergence is unchanged: $R = 1$.

Building a new series by combining known ones

14 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.
$\frac{1}{1-x} + \frac{1}{1+x} = \frac{(1+x)+(1-x)}{1-x^2} = \frac{2}{1-x^2}$. Adding the series: $\sum_{n=0}^{\infty} x^n + \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} [1 + (-1)^n] x^n$, which is $2x^n$ for even n and 0 for odd n: $2 + 2x^2 + 2x^4 + \cdots$.

Ratio Test with factorial and power combinations

15 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.
$\lim_{n\to\infty}\left|\frac{2^{n+1} x^{n+1}/(n+1)^2}{2^n x^n/n^2}\right| = \lim_{n\to\infty}\frac{2n^2 |x|}{(n+1)^2} = 2|x| < 1 \Rightarrow |x| < \frac{1}{2}$. $R = \frac{1}{2}$.

16 For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

At $x = \frac{1}{2}$: $\sum \frac{2^n (1/2)^n}{n^2} = \sum \frac{1}{n^2}$, converges (p-series, $p = 2$). At $x = -\frac{1}{2}$: $\sum \frac{(-1)^n}{n^2}$, converges absolutely. Interval: $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

Using a power series to approximate a definite integral

17 Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

$\tan^{-1}(0.5) \approx 0.5 - \frac{(0.5)^3}{3} + \frac{(0.5)^5}{5} = 0.5 - 0.04167 + 0.00625 \approx 0.4646$ (true value ≈ 0.4636, a close match).